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First report of black rot on apple fruit caused by *Diplodia seriata* in California

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ABSTRACT

A fruit rot was observed on apple trees (*Malus domestica*) cv. Fuji in a commercial orchard during 2015 and 2016 in San Joaquin Valley County, California. Early symptoms of fruit rot included light to dark-brown lesions (3-4cm in diameter) with defined margins and firm texture. In more advanced stages of the infection, the fruit became completely rotten and soft. Besides, the affected apple trees showed a plethora of fungal fruiting bodies on shoots initially blighted by *Erwinia amylovora*, the causal agent of fire blight. A fungal species was consistently isolated by cutting small pieces of symptomatic fruit and blighted shoots and placing them on petri dishes containing acidified potato dextrose agar (PDA).The dishes were incubated at 22°C for 7 days. Fungal pure cultures were obtained by transferring hyphal tips on new PDA dishes. Upon isolation, 74.5% of the colonies formed on PDA showed dense white aerial mycelium that became dark gray with age. The reverse of the colony ranged from olive green to black. The fungus formed pycnidia on pistachio leaf agar medium after 15 days of incubation at room temperature (22-25ºC). Conidia were initially hyaline, becoming dark brown at maturity, aseptate, ovoid, rounded at both ends, and 8 to 12 × 14 to 24 µm (n=50). Based on cultural and morphological characteristics the pathogen was identified as *Diplodia seriata* De Not (Phillips et al. 2007; Sutton 2014).In addition, a second species in the *Botryosphaeriaceae* was sporadically (9.5%) isolated from the blighted shoots but not from the rotted fruit. This second species was identified based on cultural and conidial characteristics as *D. mutila* (Fr.:Fr.). To confirm the morphological identification of both Botryosphaeriaceae species, the ITS and EF1-α loci were used for molecular identification with the primer sets ITS1/ITS4 (White et al. 1990) and EF1-728F/EF1-986R (Carbone et al. 1999), respectively. BLAST searches against GenBank showed ≥99% identity with reference sequences of *D. seriata* (2K33) and *D. mutila* (CBS112553) for both DNA regions (accessions nos. KF778795, KF778985 and AY259093, AY573219, respectively). The ITS and EF sequences of one representative isolate of *D. seriata* (KARE1701) and another isolate of *D. mutila* (KARE1696) were deposited in the GenBank (accessions nos. MF346337, MF346339, MF346338, MF346340 for ITS and EF1-α respectively). The pathogenicity of one representative isolate of each fungal species was tested on apple fruit cv. Fuji. For that, 20 fruit were surface-disinfected with 75% ethanol solution for 30 s. Each fruit was wounded on opposite sides using a sterile 5 mm diameter cork borer and inoculated with 5 mm diameter mycelial plugs from7-day-old PDA cultures. Ten apple fruits were inoculated with sterile PDA plugs and used as negative control. Fruits were incubated in humid chambers at 22°C for 10 days. The experiment was repeated once. Symptoms of black rot developed on all inoculated fruit showing lesions ranging from 2.1 to 6.9 cm in diameter after 10 days of

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incubation. No symptoms were observed on the negative control. Koch's postulates were fulfilled by re-isolating each fungus from the inoculated fruit. Both *D. seriata* and *D. mutila* caused similar (T-student Test, P =0.824) severity of symptoms on the inoculated apple fruit. Although black rot is well-known in the eastern United States, to our knowledge, this is the first report of black rot of apple caused by *D. seriata* in California.

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